

Operating Systems Lab Manual - Test Cases

Prime Numbers between M and N

Test Case ID	Input / Scenario	Expected Result
TC01	M=1, N=10	2,3,5,7 displayed
TC02	M=20, N=10	No output / validation message
TC03	M=11, N=11	11 displayed

Reverse and Check Palindrome

Test Case ID	Input / Scenario	Expected Result
TC01	121	Palindrome
TC02	123	Not palindrome
TC03	1200	Not palindrome

Sum of Digits

Test Case ID	Input / Scenario	Expected Result
TC01	1234	10
TC02	999	27
TC03	0	0

Linux Commands using Case

Test Case ID	Input / Scenario	Expected Result
TC01	Valid option	Correct command executed
TC02	Invalid option	Invalid choice displayed
TC03	Exit option	Program terminates

Files with RWX Permissions

Test Case ID	Input / Scenario	Expected Result
TC01	File with rwx	Filename displayed
TC02	File without execute permission	Not displayed

TC03	No matching files	No filenames shown
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Copy File within Current Directory

Test Case ID	Input / Scenario	Expected Result
TC01	Valid source and destination	File copied
TC02	Source file missing	Error message
TC03	Destination exists	Overwrite warning/error

Copy File Between Directories

Test Case ID	Input / Scenario	Expected Result
TC01	Valid directories	File copied successfully
TC02	Invalid source directory	Error message
TC03	Destination missing	Error message

Compare Two Data Files

Test Case ID	Input / Scenario	Expected Result
TC01	Common lines exist	Common lines displayed
TC02	No common lines	No common output
TC03	Unsorted files	Incorrect/validation issue

Count Vowels in String

Test Case ID	Input / Scenario	Expected Result
TC01	Hello World	3 vowels
TC02	AEIOU	5 vowels
TC03	bcdfg	0 vowels

Convert Case

Test Case ID	Input / Scenario	Expected Result
TC01	hello	HELLO
TC02	WORLD	world
TC03	TeSt	tEsT conversion appropriate

Pattern Matching in File

Test Case ID	Input / Scenario	Expected Result
TC01	Word exists	Matching lines shown
TC02	Word absent	No matches
TC03	Case-insensitive search	Matches regardless of case

Factorial of Number

Test Case ID	Input / Scenario	Expected Result
TC01	5	120
TC02	0	1
TC03	-5	Error message

Zombie and Orphan Processes

Test Case ID	Input / Scenario	Expected Result
TC01	Zombie process script	Z state visible
TC02	Orphan process script	Adopted by init
TC03	Parent terminates	Child continues running